Part I

Basic Properties of the Natural Logarithm:

Summary

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Basic properties of the natural logarithm:

- $\ln(yz) = \ln(y) + \ln(z)$
- $\ln\left(\dfrac{y}{z}\right) = \ln(y) - \ln(z)$
- $\ln(y^z) = z\ln(y)$
- For any function g: $\ln(e^{g(x)}) = g(x)$
- $\dfrac{d}{dx}(\ln(f(x)) = \dfrac{f'(x)}{f(x)}$.

Part II

Recitation Questions

Problem 1

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Problem 1 (a) *Write as an exponential with base* 5. 7^{3x}.

Explanation. $7 = 5^{\log_5(7)}$, so

$$7^{3x} = \left(5^{\log_5(7)}\right)^{3x} = 5^{3x\log_5(7)}$$

(b) *Write in terms of the natural logarithm.* $\log_3(4)$.

Explanation. The change of base formula says $\log_a(u) = \dfrac{\log_b(u)}{\log_b(a)}$, so

$$\log_3(4) = \frac{\ln(4)}{\ln(3)}$$

(c) *Expand the following:* $\log_{1/2}\left(\dfrac{6x^5(2+\tan(x))^x}{\sqrt[5]{e^{4x}+1}}\right)$.

Explanation.

$$\log_{1/2}\left(\frac{6x^5(2+\tan(x))^x}{\sqrt[5]{e^{4x}+1}}\right) = \log_{1/2}(6)+5\log_{1/2}(x)+x\log_{1/2}(2+\tan(x))-\frac{1}{5}\log_{1/2}(e^{4x}+1)$$

Problem 2

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Problem 2 *Find all real numbers x which satisfy each of the following equations.*

(a) $\log_x(25) = 2$.

Explanation. Recall that, by definition of the inverse to an exponential function,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $\log_x(25) = 2 \iff x^2 = 25$. Therefore

$$\begin{aligned} x^2 = 25 &\implies x = \pm 5, \\ &\implies x = 5 \quad \text{(base of log is always} > 0). \end{aligned}$$

Therefore $x = 5$ is the only solution to $\log_x(25) = 2$.

(b) $7^x = 15$

Explanation. Similar to the previous problem,

$$\log_b(x) = y \iff x = b^y.$$

Using this relationship we have $7^x = 15 \iff x = \log_7(15)$. Therefore $x = \log_7(15)$ is the only solution to $7^x = 15$.

(c) $\ln(x) + 1 = 0$.

Explanation. Similar to the previous two problems we'll use the relationship

$$\ln(x) = y \iff x = e^y.$$

Before applying this relationship we perform a bit of algebra first:

$$\ln(x) + 1 = 0 \implies \ln(x) = -1.$$

Now we have $\ln(x) = -1 \iff x = e^{-1}$. Therefore $x = e^{-1}(= 1/e)$ is the only solution to $\ln(x) + 1 = 0$.

Problem 3

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Problem 3 *True or False:*

(1) *If* $f(x) = (x-2)^x$, *then* $f'(x) = x(x-2)^{x-1}$.

Explanation. False. Any time that you have a function of x raised to a function of x, in order to compute the derivative you need to use logarithmic differentiation (or something equivalent).

Correct derivative of f:

$$\begin{aligned} f(x) = (x-2)^x &\implies f(x) = e^{x\ln(x-2)} \\ &\implies f'(x) = e^{x\ln(x-2)} \cdot \left(1 \cdot \ln(x-2) + x \cdot \frac{1}{x-2}\right) \\ &\implies f'(x) = (x-2)^x \cdot \left(\ln(x-2) + \frac{x}{x-2}\right) \end{aligned}$$

(2) *If* $f(x) = (3x)^x$, *then* $f'(x) = (3x)^x \ln(3x)$.

Explanation. False. Same as part (1).

Correct derivative of f:

$$\begin{aligned} f(x) = (3x)^x &\implies f(x) = e^{x\cdot\ln(3x)} \\ &\implies f'(x) = e^{x\cdot\ln(3x)} \left(1 \cdot \ln(3x) + x \cdot \frac{3}{3x}\right) \\ &\implies f'(x) = (3x)^x \left(\ln(3x) + 1\right) \end{aligned}$$

Problem 4

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Problem 4 *Find the derivatives of the following functions:*

(a) $f(x) = x^{e^x} + 7x$

Explanation. $f'(x) = \frac{d}{dx}\left(x^{e^x}\right) + \frac{d}{dx}(7x) = \frac{d}{dx}\left(x^{e^x}\right) + 7$. So the real problem is to find $\frac{d}{dx}\left(x^{e^x}\right)$.

We will use logarithmic differentiation.

First, we take natural logarithm of both sides of the equation $g(x) = x^{e^x}$.
$\ln(g(x)) = e^x \ln(x)$.
Now, we differentiate both sides.
$\frac{g'(x)}{g(x)} = e^x \ln(x) + \frac{e^x}{x}$.
Next, we solve for $g'(x)$.

$g'(x) = g(x)\left(e^x \ln(x) + \frac{e^x}{x}\right)$.
Now, we substitute.
$g'(x) = x^{e^x}\left(e^x \ln(x) + \frac{e^x}{x}\right)$.
ALTERNATIVE APPROACH

$$\begin{aligned}
\frac{d}{dx}\left(x^{e^x}\right) &= \frac{d}{dx}\left(e^{\ln(x)^{e^x}}\right) \\
&= \frac{d}{dx}\left(e^{e^x \ln(x)}\right) \\
&= e^{e^x \ln(x)}\left(e^x \ln(x) + \frac{e^x}{x}\right) \\
&= x^{e^x}\left(e^x \ln(x) + \frac{e^x}{x}\right).
\end{aligned}$$

Thus, $f'(x) = x^{e^x}\left(e^x \ln(x) + \frac{e^x}{x}\right) + 7$.

(b) $g(x) = (\ln(x) + 9)^{\sec(x^4)}$

Explanation. We will use logarithmic differentiation.

First, we take natural logarithm .

$\ln(g(x)) = \sec(x^4)\ln(\ln(x)+9)$

Then, we differentiate.

$$\frac{g'(x)}{g(x)} = \sec(x^4)\tan(x^4)4x^3\ln(\ln(x)+9) + \frac{\sec(x^4)}{x(\ln(x)+9)}$$

We multiply by $g(x)$.

$$g'(x) = g(x)\Big(4x^3\sec(x^4)\tan(x^4)\ln(\ln(x)+9) + \frac{\sec(x^4)}{x(\ln(x)+9)}\Big)$$

We substitute.

$$g'(x) = (\ln(x)+9)^{\sec(x^4)}\Big(4x^3\sec(x^4)\tan(x^4)\ln(\ln(x)+9) + \frac{\sec(x^4)}{x(\ln(x)+9)}\Big)$$

ALTERNATIVE APPROACH

$$\begin{aligned} g'(x) &= \frac{d}{dx}\left((\ln(x)+9)^{\sec(x^4)}\right) \\ &= \frac{d}{dx}\left(e^{\sec(x^4)\ln(\ln(x)+9)}\right) \\ &= e^{\sec(x^4)\ln(\ln(x)+9)}\left(4x^3\sec(x^4)\tan(x^4)\ln(\ln(x)+9) + \sec(x^4)\frac{\frac{1}{x}}{\ln(x)+9}\right) \\ &= (\ln(x)+9)^{\sec(x^4)}\left(4x^3\sec(x^4)\tan(x^4)\ln(\ln(x)+9) + \frac{\sec(x^4)}{x(\ln(x)+9)}\right). \end{aligned}$$

(c) $f(x) = \dfrac{(x+1)^5(\sin(x)+5)^4}{(x^2+5)\sqrt{x-3}}$

Explanation.

$$\begin{aligned} \ln(f(x)) &= \ln(x+1)^5 + \ln(\sin(x)+5)^4 - \ln(x^2+5) - \ln\sqrt{x-3} \\ &= 5\ln(x+1) + 4\ln(\sin(x)+5) - \ln(x^2+5) - \frac{1}{2}\ln(x-3) \end{aligned}$$

Now we differentiate.

$$\frac{f'(x)}{f(x)} = \frac{5}{x+1} + \frac{4\cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)}$$

Then we solve for $f'(x)$.

$f'(x) = f(x)\Big(\dfrac{5}{x+1} + \dfrac{4\cos(x)}{\sin(x)+5} - \dfrac{2x}{x^2+5} - \dfrac{1}{2(x-3)}\Big)$ Then we substitute.

$$f'(x) = \frac{(x+1)^5(\sin(x)+5)^4}{(x^2+5)\sqrt{x-3}}\Big(\frac{5}{x+1} + \frac{4\cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)}\Big)$$

(d) $h(x) = \dfrac{(x^2-7)^5}{\cos^7(x^2-5)}$

Explanation. Since $h(x)$ changes sign on its domain, we write :

$$\begin{aligned}\ln|h(x)| &= \ln\left(\frac{|x^2-7|^5}{|cos^7(x^2-5)|}\right)\\ &= 5\cdot\ln(|x^2-7|) - 7\cdot\ln(|\cos(x^2-5)|)\end{aligned}$$

Differentiate both sides with respect to x, (explanation given on the next page):

$$\frac{h'(x)}{h(x)} = 5\frac{2x}{x^2-7} - 7\frac{-\sin(x^2-5)2x}{\cos(x^2-5)}$$

$$h'(x) = h(x)\big(5\frac{2x}{x^2-7} - 7\frac{-\sin(x^2-5)2x}{\cos(x^2-5)}\big) =$$

$$= \frac{(x^2-7)^5}{\cos^7(x^2-5)}\big(\frac{10x}{x^2-7} + \frac{14x\sin(x^2-5)}{\cos(x^2-5)}\big) =$$

$$= \frac{(x^2-7)^4}{\cos^7(x^2-5)}\big(10x + 14x(x^2-7)\tan(x^2-5)\big)$$

EXPLANATION: The function $h(x)$ assumes negative values on some intervals. On those intervals

$$\big(\ln(|h(x)|)\big)' = \big(\ln(-h(x))\big)' = \frac{-h'(x)}{-h(x)} = \frac{h'(x)}{h(x)},$$

Also, e.g., if $\quad x^2-7<0$

$$\big(\ln(|x^2-7|)\big)' = \big(\ln(-(x^2-7))\big)' = \frac{-2x}{-(x^2-7)} = \frac{2x}{x^2-7}\ .$$